From Heterogeneous Visual Cues to Bounded Grasp Decisions for Upright Vessel Instances

Matthäus Horn
dept. Information Technology
Salzburg University of Applied Sciences
Salzburg, Austria
ORCID: 0000-0002-1166-9629

***Abstract*—Side grasping an upright cup, glass, or mug does not always require a complete six-degree-of-freedom pose aligned to a computer-aided design model. Instead, the robot must decide whether the available visual evidence supports an executable external side grasp. This paper presents a reduced grasp-oriented vessel belief containing center, approximate upright axis, object height, grasp height, local grasp width, object family, cue validity, and an analytical grasp-readiness risk indicator. An overhead Intel RealSense D455 provides global color and depth context and candidate proposals from a YOLO11n pose detector. During approach, an eye-in-hand Intel RealSense D405 refines local grasp geometry using depth profiles and color-image flank and silhouette cues. Segment Anything Model 2 contours, table constraints, and temporal consistency provide complementary evidence. Explicit cue arbitration maps the belief to a pre-grasp, one predefined additional observation, or blocking. The local grasp module uses no object-specific computer-aided design model or stored grasp template. Two evaluations of 180 trials each were performed in a non-optimized corridor under different daylight and shadow conditions. Across 360 trials, 335 upright candidates reached the vessel stage, 222 intended targets were grasped, and 235 physical grasps succeeded. Candidate availability increased from 160 of 180 to 175 of 180 trials, while target success conditioned on eligibility remained similar at 66.9 and 65.7 percent. Three detector-held-out instances achieved 88 target successes in 130 planned trials and were not systematically weaker than detector-seen instances. The results demonstrate an inspectable interface between partly unreliable vessel perception and bounded robot execution.**

***Index Terms*—robot grasping, task-oriented representation, color and depth perception, cue arbitration, transparent objects**

I. Introduction

Learned grasping methods can generate broad grasp proposals from point clouds, synthetic data, and volumetric scene representations [1]–[3]. A complete manipulation system must still transform those perceptual outputs into a bounded physical action. This transition is difficult when depth is sparse, transparent or glossy contours depend on the background, neighboring objects contaminate local geometry, or an upstream detector selects a plausible but incorrect target.

Upright vessel-like objects provide a useful restricted setting. Cups, glasses, and mugs vary in material, taper, rim geometry, and handle structure, yet a parallel-jaw side grasp depends mainly on object center, vertical extent, grasp height, local width, and approximate axis. A complete CAD-aligned pose can therefore be unnecessary. Transparent-object methods commonly recover richer intermediate representations through depth completion, keypoints, neural geometry, or full pose estimation [4]–[8]. This work addresses a narrower execution question: given a coarse upright vessel candidate, are the available observations sufficient for a side grasp, should the robot observe once more, or should execution be blocked? Because such vessels may contain food or beverages, the system deliberately targets external side grasps and avoids inserting the gripper into the vessel interior.

The approach is inspired by uncertainty-aware pre-grasp planning. Naik et al. maintain multimodal 6D pose distributions and use their uncertainty during mobile manipulation [9], [10]; related work explicitly links pre-grasp approach behavior to the expected value of improved observations [11]. We transfer this principle to a smaller task state. The proposed system does not require a CAD model, a renderer, or a complete pose distribution. Instead, it retains only grasp-relevant vessel variables and excludes cues that are unstable, shifted, inconsistent, or physically infeasible.

The paper makes three contributions. First, it defines a reduced grasp-oriented vessel belief for upright instances and instantiates it from overhead context, eye-in-hand D405 depth and RGB geometric cues, promptable SAM2 masks, and table constraints. Second, it presents an execution-aware arbitration policy that maps the belief to a pre-grasp, a predefined observation route, or blocking. Third, it reports 360 planned robot trials, including detector-held-out instances and six-object mixed clutter, using metrics that separate candidate availability, expected-target selection, intended-target success, physical grasp success, wrong-object grasps, and unsafe motion failures.

II. Related Work

A. Unknown-object grasping and task representations

Point-cloud grasp pose detection [1], synthetic robustness analysis in Dex-Net [2], and volumetric grasp quality prediction in clutter [3] aim to produce grasp candidates over broad object sets. Our representation is deliberately narrower. The proposed method starts from a selected upright vessel candidate. Rather than generating arbitrary grasp poses, it estimates only the geometric and validity variables required to determine whether

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an external side grasp can be executed. In this paper, *instance-agnostic* refers specifically to the local grasp module, which uses no instance-specific CAD model, stored pose template, or pre-recorded grasp.

B. Transparent and specular perception

Existing transparent-object methods address depth completion and related real-data benchmarks [4], [6], keypoint estimation [5], multiview grasp geometry [7], and full pose estimation [8]. These methods recover geometry or pose more directly than the proposed system. Here, transparent and glossy objects are difficult members of a wider vessel family. Partial cues are arbitrated for grasp execution rather than fused into a claim of complete shape reconstruction.

C. Multi-view uncertainty and pre-grasp observation

Naik et al. show that multiple pose hypotheses and camera-view reliability can support pre-grasp decisions [9], [10]. The proposed vessel belief transfers that idea to a reduced state over center, axis, height, grasp height, width, family, cue mode, and grasp-readiness risk. The observe action is requested when evidence is plausible but unstable. Its trajectory is predefined and is not a general next-best-view optimizer.

III. System and Method

Fig. 1 summarizes the complete global-to-local pipeline and the bounded action space.

A. Hardware, calibration, and global proposal

The system consists of a UR5e manipulator, a Robotiq Hand-E parallel gripper, an overhead Intel RealSense D455 RGB-D camera, and an eye-in-hand Intel RealSense D405. A ChArUco target [12] was used for camera calibration. The overhead camera-to-base transform combined rigid alignment of manually touched board points with the visual board pose. The eye-in-hand transform was estimated from 24 robot poses using OpenCV hand-eye calibration with the Tsai method. Intrinsic calibration RMS errors were 0.184 pixels for the D455 and 0.191 pixels for the D405. A shared-board validation pose showed a 0.895 mm translational discrepancy, reported only as a local alignment check.

The D455 provides global table context, candidate proposals, and deterministic target ordering. An Ultralytics YOLO11n pose model [13] was trained for cup, glass, and mug, each labeled as upright, upside down, or lying. It returns a bounding box and one center keypoint. Final inference used 960-pixel input, confidence 0.50, IoU threshold 0.35, and at most 20 detections; only upright candidates enter the vessel stage.

The proposal dataset contained 445 images, with 356 training and 89 validation images and 13 empty negatives. The random image-level split contains temporally adjacent frames and repeated physical instances, so detector validation is treated as an implementation check rather than evidence of instance-independent generalization. The detector was trained on multiple physical instances of each vessel family.

In clutter, candidates are processed in a deterministic row-major order. Starting from the row closest to the robot as viewed from the front of the workspace, the system proceeds from left to right before continuing with the next row farther back. The expected target is defined by this ordering. A target lock is maintained during approach so that local refinement remains associated with the selected proposal.

B. Local D405 Geometric Cues and SAM2

The D405 refinement stage combines local depth geometry with image-based cues from the same eye-in-hand camera. Valid depth points above the table plane are transformed into a local vessel frame and sampled in horizontal profile bands. Within these bands, object-centered flank fitting estimates local width, center consistency, and side-wall support. In parallel, RGB row profiles provide two-dimensional flank hypotheses, an outer-silhouette estimate at the intended grasp height, and additional evidence for the rim or opening region. A valid two-dimensional profile can supplement or replace a weak depth profile, while the rim estimate combines compatible three-dimensional and image-based observations. The center-out formulation limits clutter inflation that could occur if neighboring points were included in a single outer hull. Profiles are rejected when their centers are shifted, their flanks are unstable, their widths are inflated by clutter, or their support near the intended grasp height is insufficient.

SAM2 [14] contributes a prompt-based mask, contour, center, and image-space width hypothesis. It is an auxiliary cue rather than a replacement for local geometry. A SAM2 cue can be excluded when its score is weak, the mask collapses, the contour is implausibly broad, the center shifts away from the locked target, or its width disagrees strongly with D405 depth and RGB geometric cues. Excluded cues remain available for diagnosis but cannot trigger motion. Each SAM2 snapshot is inferred independently from the current D405 RGB image and prompt; no temporal mask propagation or video memory is used. Temporal consistency is evaluated on the resulting geometric cues rather than within SAM2 itself.

C. Reduced vessel belief

For an eligible upright candidate, the system maintains

$$b_v = (\mathbf{c}, \mathbf{a}, h, h_g, w_g, f, q, r), \tag{1}$$

where $\mathbf{c}$ is the center, $\mathbf{a}$ an approximate upright axis, h the object height, h_g the grasp height, w_g the local grasp width, f the family, q the cue-validity mode, and $r \in [0, 1]$ an analytical grasp-readiness risk score. Fig. 2 illustrates the retained state variables and their geometric interpretation for an external side grasp.

The implementation evaluates discrete hypotheses formed from available center, height, width, ring, taper, family, and cue-source alternatives. Weights combine source priors, cue quality, measurement consistency, and physical feasibility. The normalized hypotheses provide a mean, spread, entropy, effective sample size, and dominant alternatives. These are diagnostic summaries rather than calibrated probabilities. The cue mode q records which source configuration is usable, weak, excluded, or dominant. The grasp-readiness risk $r \in [0, 1]$ is

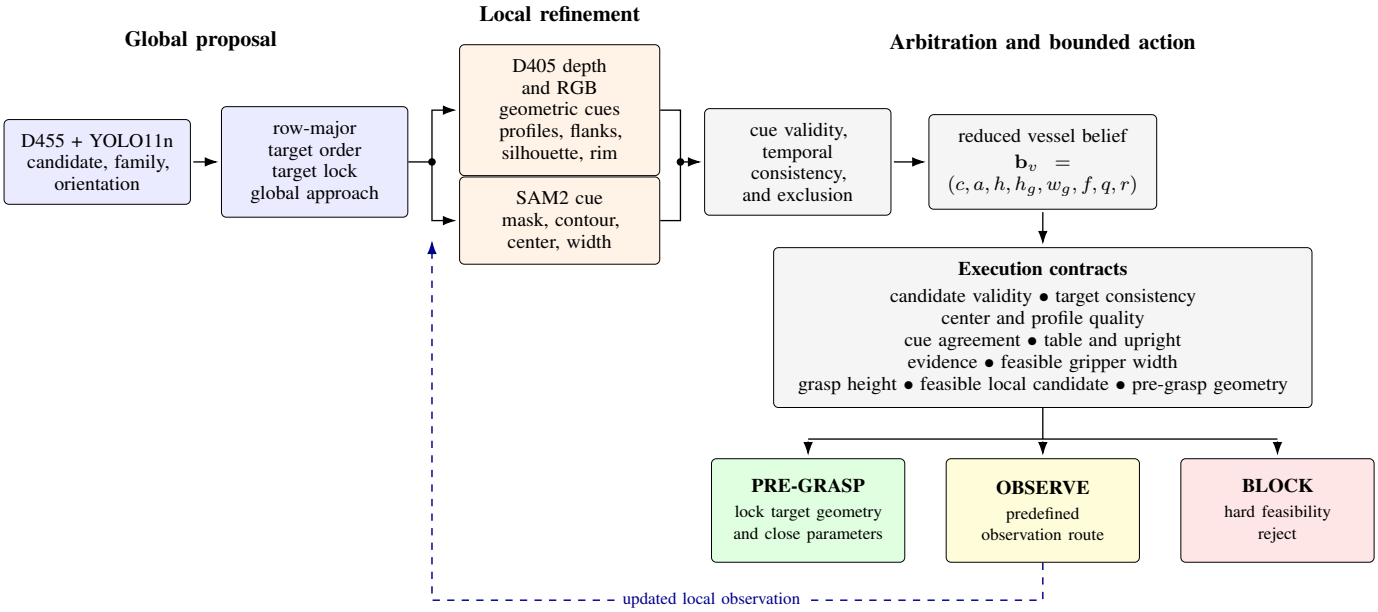


Fig. 1. System pipeline. D455 and YOLO11n provide an upright candidate and deterministic target order. Eye-in-hand D405 and SAM2 refine local geometry and cue validity. Execution contracts map the reduced vessel belief to PRE-GRASP, OBSERVE, or BLOCK.

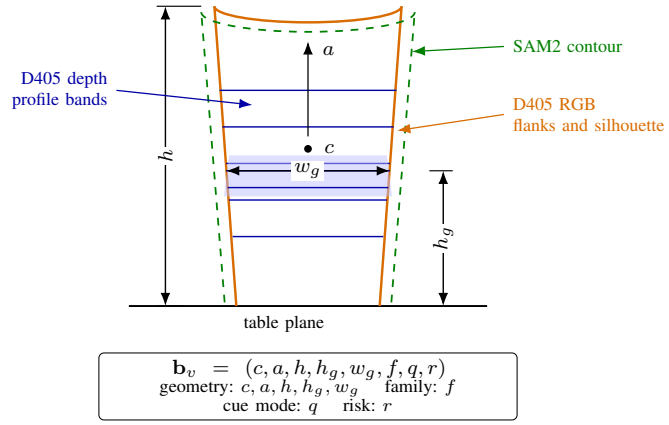


Fig. 2. Reduced grasp-oriented vessel belief. D405 depth and RGB cues, the SAM2 contour, and the table plane provide complementary evidence for an external side grasp. Variables f, q, and r denote vessel family, cue-validity mode, and grasp-readiness risk. Colors distinguish cue types; the schematic is not to scale.

a fixed analytical score combining normalized entropy, width spread, grasp-height spread, gripper feasibility, table evidence, upright evidence, multimodality, and the absence of a feasible local grasp candidate with weights 0.22, 0.22, 0.14, 0.18, 0.10, 0.08, 0.06, and 0.12, respectively. Width and grasp-height spreads are normalized by 10 mm and 14 mm, respectively, with their individual risk contributions capped at one. Risk is classified as low for $r < 0.34$, medium for $0.34 \leq r < 0.67$, and high otherwise. The score is heuristic, not learned, and is not a calibrated probability of grasp failure.

D. Execution contracts and bounded action

The fused vessel belief, a feasible local grasp candidate, and the committed pre-grasp geometry are distinct. A feasible local candidate is one concrete center, grasp-height, and width hypothesis satisfying the local geometric and gripper constraints. Only after the remaining target-lock, consistency, and execution-contract checks does its geometry become the committed pre-grasp state. Before pre-grasp motion, the system checks candidate validity, expected-target and target-lock consistency, center tolerance, profile quality, cue agreement, feasible gripper width, minimum grasp height, feasible local candidate construction, vessel-belief validity, and pre-grasp geometry. Plausible but unstable evidence triggers the predefined observation route, whereas failed hard geometric or gripper-feasibility contracts block execution. Only a valid and stable belief proceeds to the pre-grasp motion. Once this motion is initiated, the selected target geometry and close parameters are locked for the remaining grasp sequence. The subsequent close command uses this stored commitment rather than re-estimating the partly occluded object from the post-approach camera view. The primary execution and observation thresholds used in the final evaluation are summarized in Table I.

TABLE I
PRIMARY EXECUTION-CONTRACT AND OBSERVE THRESHOLDS.

Stage	Final thresholds
Belief gate	readiness ≥ 0.72, commit ≥ 0.70, ≥ 4 stable frames, $\sigma_{xy} \leq 7.5$ mm, consistency ≤ 10 mm
Selected state	confidence ≥ 0.50, $p_g \geq 0.75$, $h_g \geq 60$ mm, commit width 30–80 mm
Fallback	score ≥ 0.50, width 45–82 mm, center shift ≤ 8 mm, D405 difference ≤ 6 mm
OBSERVE	≤ 1 cycle, 20° arc, lateral shift ≤ 30 mm, dwell 0.55 s

The 60 mm grasp-height limit is a measured table-clearance guard; the remaining thresholds were fixed empirically during system development and were not optimized on the reported

outcomes.

IV. Experimental Protocol

A. Objects, scenes, and environmental conditions

Each evaluation contained 180 planned trials: 30 cup single, 30 glass single, 30 mug single, 30 cup same-family clutter, 15 glass same-family clutter, 15 mug same-family clutter, and 30 mixed-clutter trials. Each mixed-clutter scene contained six objects, comprising one detector-seen and one detector-held-out instance from each of the three vessel families. Across the 30 mixed-clutter trials, the deterministic ordering produced ten expected cup targets, ten expected glass targets, and ten expected mug targets.

The evaluation used three physical instances per vessel family. CUP03, GLASS02, and MUG03 were excluded from detector training and served as held-out instances during evaluation. Among the detector-seen evaluation objects, CUP01, GLASS01, and MUG01 had repeatedly challenged the system during development, whereas CUP02, GLASS03, and MUG02 had performed comparatively reliably. These labels describe prior development experience rather than separate benchmark classes. Object heights ranged from 81.9 to 134.0 mm and local diameters from 60.5 to 81.8 mm. MUG01 had a local diameter of 81.8 mm, close to the admissible upper width range, making it especially sensitive to positive width bias and feasibility rejection. Mug handles were oriented rearward to minimize direct visibility and interference.

The same nominal positions and deterministic target-order rule were used in both evaluations. Positions P1–P5 denoted similar table regions, but objects were placed manually without fixtures. Trial-to-trial placement could therefore vary by a few centimeters in the table plane. This introduces small spatial perturbations while preventing the two runs from being treated as exact paired repetitions.

Trials were conducted in a non-optimized corridor beside a window. Evaluation 1 spanned two days with varying daylight, window shadows, and occasional artificial lighting, whereas Evaluation 2 was conducted during one afternoon and evening under more uniform mixed daylight. The differing illumination profiles preclude causal attribution of cross-run changes to diagnostic load.

Evaluation 1 enabled dense logging and visualization. Evaluation 2 retained the same perception, belief, motion, and gripper logic while disabling nonessential recording and visualization; sparse decision, cue-state, runtime, YOLO, SAM2, and D405 diagnostics remained active.

B. Outcome definitions and data audit

Let P denote planned trials and E trials in which an upright candidate reached the vessel stage. Sel denotes correct selection of the expected target. Target success TSucc requires selection and successful grasp of that target. Physical grasp success GSucc also counts successful grasps of a wrong object. Wrong counts those physically successful wrong-object grasps, and U counts unsafe failures. Manual trial labels are the primary outcome source for both evaluations. Automatic traces are used only to audit state transitions and motion completion because they cannot determine whether the intended object was held. A standardized mutually exclusive stage-level failure taxonomy was available only for Evaluation 2 because only this run recorded the Fast Decision Trace format required for complete trial-level state-sequence matching.

All 175 valid trials in Evaluation 2 were matched to a trace; the five invalid trials had no started trace. Manual labels were cross-checked against the available traces, and one internally inconsistent record was corrected before aggregation. Missing technical fields were not imputed. Because this is a fixed systems protocol rather than random population sampling, the analysis emphasizes absolute counts, planned-trial rates, and rates conditioned on eligibility. Wilson 95% intervals are additionally reported descriptively for the main success proportions and are not interpreted as population-level inference.

V. Results

A. Aggregate performance and stage separation

Across both evaluations, 335 of 360 planned trials reached the vessel stage, 304 selected the expected target, 222 completed an intended-target grasp, and 235 completed a physical grasp. Evaluation 1 produced $E = 160$, $Sel = 141$, $\mathrm{TSucc} = 107$, $\mathrm{GSucc} = 115$, $\mathrm{Wrong} = 8$, and $U = 3$. Evaluation 2 produced $E = 175$, $Sel = 163$, $\mathrm{TSucc} = 115$, $\mathrm{GSucc} = 120$, $\mathrm{Wrong} = 5$, and $U = 6$. Table II separates proposal availability and target selection from end-to-end and eligibility-conditioned grasp outcomes.

Candidate availability increased from 88.9 to 97.2 percent, and selection accuracy among eligible candidates increased from 88.1 to 93.1 percent. End-to-end target success increased from 59.4 to 63.9 percent, while physical grasp success increased from 63.9 to 66.7 percent. Conditioned on eligibility, target success remained close at 66.9 and 65.7 percent, and physical grasp success was 71.9 and 68.6 percent. The repeated run therefore reproduced similar downstream performance while increasing the number of trials that reached the local belief stage. OBSERVE was triggered in 82/160 eligible trials in Evaluation 1 and 84/175 in Evaluation 2. After the additional observation, 47/82 and 44/84 trials, respectively, progressed to PRE-GRASP. Among these progressed cases, intended-target success was 38/47 (80.9%) and 36/44 (81.8%), while physical grasp success was 41/47 (87.2%) and 38/44 (86.4%). OBSERVE was triggered only from states that had not yet been released for PRE-GRASP. These conditional outcomes characterize post-observation progression but do not establish a causal benefit because no matched no-OBSERVE control condition was evaluated.

For eligibility-conditioned target success, descriptive 95% Wilson intervals were 59.3–73.7% and 58.4–72.3% for Evaluations 1 and 2; the corresponding physical-grasp intervals were 64.5–78.3% and 61.4–75.0%. Dense per-trial grasp-readiness risk values were retained only for Evaluation 1 because the Fast Eval configuration disabled the dense quality trace while leaving the belief and risk computations active.

TABLE II

AGGREGATE PERFORMANCE ACROSS BOTH EVALUATIONS. COUNTS ARE GIVEN IN PARENTHESES; Δ DENOTES EVALUATION 2 MINUS EVALUATION 1 IN PERCENTAGE POINTS.

Metric	Evaluation 1	Evaluation 2	Δ [pp]
Proposal and selection			
Candidate availability	88.9% (160/180)	97.2% (175/180)	+8.3
Selection accuracy	88.1% (141/160)	93.1% (163/175)	+5.0
Grasp outcomes			
Target success, planned trials	59.4% (107/180)	63.9% (115/180)	+4.5
Target success, eligible trials	66.9% (107/160)	65.7% (115/175)	−1.2
Physical grasp success, planned trials	63.9% (115/180)	66.7% (120/180)	+2.8
Physical grasp success, eligible trials	71.9% (115/160)	68.6% (120/175)	−3.3

Planned trials include all 180 trials in each evaluation. Eligible trials are those in which an upright candidate reached the vessel stage. Target success requires a successful grasp of the expected target. Physical grasp success additionally includes successful grasps of a different object.

TABLE III

BLOCK-LEVEL OUTCOMES. P DENOTES PLANNED TRIALS AND E ELIGIBLE TRIALS. TARGET SUCCESS REQUIRES A SUCCESSFUL GRASP OF THE EXPECTED TARGET. PHYSICAL SUCCESS ALSO INCLUDES SUCCESSFUL GRASPS OF ANOTHER OBJECT.

		Evaluation 1				Evaluation 2			
Block	P	E	Target	Physical	Wrong	E	Target	Physical	Wrong
Cup single	30	30	27	27	0	30	22	22	0
Glass single	30	15	15	15	0	27	19	19	0
Mug single	30	27	13	13	0	30	18	18	0
Cup clutter	30	30	23	23	0	30	24	24	0
Glass clutter	15	13	6	8	2	14	6	9	3
Mug clutter	15	15	3	5	2	15	8	9	1
Mixed clutter	30	30	20	24	4	29	18	19	1
All	180	160	107	115	8	175	115	120	5

Across all 160 eligible Evaluation 1 trials, median final grasp-readiness risk was 0.210 for intended-target successes and 0.264 for target failures; the corresponding medians for physical grasp success and failure were 0.215 and 0.264. The three unsafe motion failures had low risk scores of 0.192, 0.265, and 0.270, confirming that r captures local perception and grasp feasibility rather than trajectory-level collision risk.

The difference between target and physical success remained important. Evaluation 1 contained eight successful grasps of a wrong object; Evaluation 2 contained five. A physical-only metric would count these 13 grasps as successes and obscure target-order or target-lock errors.

Evaluation 1 permits a coarser stage audit: 20/180 trials did not reach eligibility, 19/160 eligible trials selected the wrong expected target, and 34/141 correctly selected trials did not achieve target success. Three unsafe motion failures were identified manually. For Evaluation 2, the standardized traces provide a mutually exclusive stage-level accounting of all 180 planned trials: 115 target successes, 5 trials without an eligible upright candidate, 12 wrong-target selections, 37 correctly selected trials blocked before PRE-GRASP, 2 PRE-GRASP sequences that did not reach gripper closing, 3 closing attempts without grasp success, and 6 unsafe table-contact failures.

The block-level outcomes are discussed next and are summarized in Table III.

B. Instance coverage and scene complexity

Detector-held-out instances were not systematically weaker than detector-seen instances. Across both evaluations, CUP03, GLASS02, and MUG03 reached eligibility in 124/130 planned trials and achieved 88 target successes and 91 physical grasps. The six detector-seen instances achieved 134 target successes in 230 planned trials. These counts do not isolate detector generalization because object geometry and scene allocation differ, but they show that the local instance-agnostic grasp module remained usable on the held-out vessels.

The block-level outcomes separate candidate availability, intended-target success, physical grasp success, and wrong-object grasps. Unsafe motion failures are analyzed separately in Section V-C because they represent motion-planning failures rather than grasp-outcome categories. Glass single candidate availability rose from 15/30 to 27/30, indicating that the weak first-run end-to-end glass result was largely upstream. Same-family mug clutter improved from 3/15 to 8/15 target successes. Mixed clutter contained six objects, including three held-out instances. Across both runs it produced 38/60 target successes and 43/60 physical grasps. Correct selection improved in the second run and wrong-object grasps fell from four to one, although physical success decreased from 24/30 to 19/30.

MUG01 illustrates the interaction between physical geometry and conservative feasibility. It reached the vessel stage in 19/20 trials in each evaluation, yet target success was 9/20 and 7/20. With a measured diameter of approximately 81.8 mm, MUG01 lies close to the upper feasible grasp-width range. A moderate width overestimation can therefore trigger the feasibility gate more readily than for narrower objects. This conservative behavior favors blocking over attempting a poorly conditioned close near the gripper limit.

C. Runtime indicators and motion failures

Sparse Evaluation 2 traces yielded median rate indicators of approximately 1.33 frames per second for single-object trials, 0.72 for same-family clutter, and 0.60 for mixed clutter. Median D455 inference time increased from approximately 50 ms in single scenes to approximately 140 ms in clutter. These periodic samples indicate a scene-complexity trend but do not constitute a complete frame-time benchmark. The lightweight diagnostic configuration nevertheless retained all cue computations. The dense Evaluation 1 logs additionally provided pre-approach SAM2 inference times for 146 of 160 eligible trials, with a median of 389 ms and an interquartile range of 229–467 ms.

Evaluation 1 contained three unsafe failures. One occurred at nominal position P5, where an elbow joint contacted the table near the right workspace boundary. The other two occurred at positions P2 and P4. In these trials, a vessel in the rear row was selected and the grasp motion was executed despite another object occupying the approach path, resulting in object collisions. All six unsafe failures in Evaluation 2 occurred at P5

and involved the same table-contact failure mode observed there in Evaluation 1. Manual placement without fixtures could shift objects by a few centimeters beyond their nominal positions. Together, these failures expose two distinct motion-planning limitations: insufficient collision clearance near the workspace boundary and missing obstacle awareness along the approach path to rear-row targets. They should therefore be interpreted as motion failures rather than failures of the local vessel perception.

VI. Discussion

The experiments support the central systems claim that a reduced grasp-oriented state can provide an inspectable interface between heterogeneous perception and robot execution. Similar eligibility-conditioned target success across two environmental realizations, despite changing candidate availability, separates limitations in proposal, selection, local grasp geometry, and robot motion and prevents wrong-object grasps from being reported as task success.

The held-out-object results strengthen the instance-agnostic interpretation of the local module. The three detector-held-out vessels achieved high eligibility and were not systematically weaker than detector-seen instances. This does not establish open-world detection, because the full pipeline still begins with a supervised family and orientation classifier. It does show that the D405 depth and RGB geometric cues, SAM2 cue, reduced vessel state, and execution contracts did not require object-specific CAD models or stored grasp templates.

The method also provides an explicit role for conservative rejection. Cue exclusion prevents one plausible frame from directly commanding the gripper, while the predefined observation route allows uncertain geometry to be reconsidered. The MUG01 results show the cost of a narrow feasibility margin, while the table and object collisions show that perceptual validation cannot substitute for trajectory planning with collision avoidance and obstacle awareness. Both limitations are exposed by the stage-separated evaluation.

The experiments validate the integrated pipeline but neither compare it against matched baselines nor isolate individual cue contributions, because no matched baseline, cue-ablation, or no-OBSERVE experiment was performed. The grasp-readiness risk is heuristic and excludes trajectory-level collision risk. Track identity remains limited in arbitrary clutter and only upright vessels are executable. Future work should address matched baselines and cue ablations, calibrated diagnostics, collision-aware planning, and explicit view selection.

VII. Conclusion

This paper presented a reduced grasp-oriented vessel belief that converts heterogeneous visual cues into bounded side-grasp decisions without requiring CAD alignment or instance-specific grasp templates. Across 360 planned trials, the system achieved 222 intended-target grasps and 235 physical grasps, while detector-held-out vessels remained usable. The repeated evaluation preserved eligibility-conditioned performance while increasing candidate availability and exposed where further engineering improvements are required.

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